C7 Oxidation, Reduction and Redox – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 1: Definitions and Oxidation State Rules **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Oxidation				Loss of electrons
2	Reduction				Gain of electrons
3	Oxidising agent				Electron acceptor
4	Reducing agent				Electron donor
5	Oxidation state of an uncombined element				0
6	Sum of oxidation states in a compound				0
7	Sum of oxidation states in an ion				Charge on the ion
8	Usual oxidation state of O				−2
9	Oxidation state of O in peroxides				−1
10	Oxidation state of O in OF_2				+2

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

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AQA A-level Chemistry • Set 2: More Oxidation State Rules **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Usual oxidation state of H				+1
2	Oxidation state of H in metal hydrides				−1
3	Oxidation state of F in compounds				−1
4	Oxidation state of Group 1 metals				+1
5	Oxidation state of Group 2 metals				+2
6	Usual oxidation state of Cl in compounds				−1
7	Oxidation state of S in SO_4^{2-}				+6
8	Oxidation state of Mn in MnO_4^-				+7
9	Oxidation state of Cr in $Cr_2O_7^{2-}$				+6
10	Oxidation state of N in NO_3^-				+5

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

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AQA A-level Chemistry • Set 3: Half-Equations and Oxidation States **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Half-equation				Equation showing electrons gained or lost by one species
2	Disproportionation				Same element oxidised and reduced
3	Half-equation for Fe^{2+} to Fe^{3+}				$Fe^{2+} \rightarrow Fe^{3+} + e^-$
4	Half-equation for Cl_2 to Cl^-				$Cl_2 + 2e^- \rightarrow 2Cl^-$
5	Oxidation state of Cl in ClO^-				+1
6	Oxidation state of Cl in ClO_3^-				+5
7	Oxidation state of N in NH_3				−3
8	Oxidation state of C in CO_2				+4
9	Oxidation state of Fe in Fe_2O_3				+3
10	Oxidation state of S in H_2S				−2

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.